Project Design Phase-I – Problem Solution Fit

Project: UCab – Cab Booking System

1. CUSTOMER SEGMENT(S) CS

- Urban commuters & working professionals (age 20–45) who travel daily.
- Students who need affordable and quick transport to college or events.
- Frequent travellers who need reliable airport and intercity rides.
- Drivers / cab owners seeking a platform to manage and earn from rides.

6. CUSTOMER CONSTRAINTS CC

- Budget sensitivity — riders expect affordable, transparent fares with no hidden charges.
- Smartphone & internet dependency — requires stable mobile connectivity for booking & tracking.
- Trust & safety concern — riders hesitant to use platforms without driver verification.
- Drivers need low commission rates and quick payout mechanisms to stay on the platform.

5. AVAILABLE SOLUTIONS AS

- Existing apps: Ola, Uber — feature-rich but complex UI, surge pricing, no student/discount tier.
- Auto-rickshaws & local taxis — offline, cash-only, no tracking, no receipts.
- Public transport — cheap but slow, infrequent, no door-to-door coverage.

UCab differentiates with real-time tracking, role-based access, transparent fare estimates, and a support ticket system.

2. JOBS-TO-BE-DONE / PROBLEMS J&P

- Riders struggle to find available, verified cabs quickly, especially during peak hours.
- No real-time tracking leads to uncertainty and safety concerns during rides.
- Opaque pricing causes distrust — riders don't know the fare until after the ride.
- Drivers lack a unified dashboard to manage ride requests, earnings, and history.
- No easy way to raise complaints or get support after a bad ride experience.

9. PROBLEM ROOT CAUSE RC

- No centralised, real-time platform connecting riders and nearby verified drivers.
- Lack of transparent fare computation before booking creates information asymmetry.
- Absence of role-based dashboards means drivers and riders use the same cluttered interface.
- Offline payment and cash dependency result in no digital record, disputes, and insecurity.

7. BEHAVIOUR BE

- Riders habitually open a cab app as soon as they need to travel — booking is impulsive and time-sensitive.
- Riders compare ETAs and fares before confirming, so upfront fare display is critical.
- Drivers prefer accepting nearby requests to minimise dead mileage and maximise earnings.
- Both riders and drivers check ride history and ratings frequently to manage reputation.

3. TRIGGERS TR

- Missing a bus or auto in peak hours → immediate need for a reliable cab.
- Urgent travel need — airport, hospital, office meeting with no time to wait.
- Safety concern at night — riders want a tracked, verified cab over an unknown vehicle.
- Recommendation from a friend or colleague who had a smooth UCab experience.

10. YOUR SOLUTION SL

UCab is a MERN stack cab booking web app that provides:

- Instant ride booking with real-time cab availability and upfront fare estimates.
- Live GPS tracking from pickup to drop-off via Socket.IO and Leaflet maps.
- Role-based dashboards for Riders, Drivers, and Admins with JWT authentication.
- Secure digital payments with automatic receipt and payment history.
- Driver rating system and a support ticket module for post-ride issues.

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE

- Booking via UCab web app — search location, choose cab, confirm.
- Real-time ride tracking on Leaflet map within browser.
- Online payment via saved method; download digital receipt.
- Rate driver and submit support ticket through the app.

8.2 OFFLINE

- Word-of-mouth referrals from colleagues, friends, university campuses.
- Driver onboarding via physical verification of licence and vehicle documents.

4. EMOTIONS: BEFORE / AFTER EM

BEFORE (problem state):

• Anxious, stressed, uncertain — "Will I get a cab? Will it arrive on time?"

• Frustrated with cash-only rides, unknown fares, and no safety tracking.

AFTER (using UCab):

• Confident, in control, and safe — track the ride at every step.

• Satisfied — transparent fare, digital receipt, easy rating. Likely to return.

SOLUTION FIT SUMMARY

UCab directly addresses every identified customer pain point: instant booking solves availability anxiety, real-time tracking resolves safety concerns, upfront fare estimates eliminate pricing distrust, and the digital payment + receipt system replaces unreliable cash transactions.

The role-based architecture (Rider / Driver / Admin) ensures each user gets a focused, clutter-free experience. The driver rating and support ticket modules build trust and loyalty, directly converting emotional pain into long-term platform engagement.

✓ Customer Segment fits Solution ✓ Constraints addressed by design ✓ Behaviour channels leveraged via web app ✓ Emotions transformed from anxiety to confidence